

Md Masud-Ul-Alam

Ph.D. candidate

University of Georgia | Skidaway Institute of Oceanography

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Research Interests

- Phytoplankton ecology, biogeography, community variability, and primary productivity
- Marine carbon cycling, cross-shelf exchange, and carbon sequestration
- Physical-biological coupling across plumes, fronts, eddies, storms, and stratified systems
- Ocean color and optical oceanography using PACE, MODIS, VIIRS, and Sentinel-3
- Multi-trophic ecosystem dynamics, water quality, and harmful bloom applications
- Integration of satellites, ships, gliders, BGC-Argo, models, machine learning, and HPC

Education

- 2021–Present **Ph.D. Candidate**, Department of Marine Sciences, University of Georgia (UGA), Athens, USA (Expected to graduate in January 2027)
- 2016–2017 **Postgraduate Education and Training in Ocean Observation**, Alfred Wegener Institute (AWI), Germany
- 2014–2015 **M.S. in Oceanography**, University of Dhaka, Bangladesh
- 2008–2012 **B.Sc. in Fisheries and Marine Resources Technology**, Khulna University, Bangladesh

Employment

- 08/2021 –Present **Graduate Research Assistant** at the Department of Marine Sciences, UGA Skidaway Institute of Oceanography, USA
- 01/2018 – 01/2019 **Scientific (Research) Officer**, Physical and Space Oceanography, Bangladesh Oceanographic Research Institute, Bangladesh
- 09/2016 – 09/2017 **Research Fellow** at Alfred Wegener Institute (AWI), Helmholtz Centre for Polar and Marine Research, Germany
- 08/2015 – 12/2015 **Internship** at National Institute of Oceanography, Goa, India
- 01/2019 –Present **Lecturer** at the Department of Oceanography and Hydrography, Bangladesh Maritime University (BMU), Bangladesh (on study leave)
- Summer 2025 **Graduate Teaching Assistant** at the Department of Marine Sciences, UGA Skidaway Institute of Oceanography, USA

Teaching Experience

At BMU (01/2019 - 07/2021)

Undergraduate Course Taught

Classroom instruction (Class size: 35-40 students):

- OCN 1101 Introduction to Oceanography (3 credits). Spring 2019, 2020.
- OCN 1203 Biological Oceanography (3 credits). Fall 2020, 2021.
- OCN 2107 Marine Resources (3 credits). Spring 2020.
- OCN 3101 Ocean Optics (3 credits). Spring 2020.

- OCN 1202 Physical Oceanography Lab (1.5 credits). Fall 2019, 2020.
- OCN 2204 Fisheries Oceanography lab (1.5 credits). Fall 2020.
- OCN 3211 Field Work III: Oceanographic Cruise Techniques (1.5 credits). Fall 2019.
- Teaching Assistant for an undergraduate course of 56 students (MARS 1011E - Intro to Marine Environment) at UGA. Summer 2025.

Certificates/Course for Teaching and Quality Assurance in Higher Education

- 2019: Designing effective quality Assurance Systems in Higher Education in Bangladesh, organized by the Centre for Innovative Leadership Navigation (CILN), London, UK. (5 days, equivalent to 25 Continuous Professional Development Points. This training focused on the outcome-based education and quality assurance in higher education.
- 2019: Foundation Training for University Teachers, Graduate Training Institute at Bangladesh Agricultural University, Bangladesh (03/12/2019 – 04/07/2019, total 150 hours). The certificate course covered Teaching Methods and Techniques, Research Methodology and Project Management, Scientific Writing, and Statistics and Data Analysis at the graduate level institutions.
- 2025: GRSC 7770 course (3 credits), as the mandatory TA training requirement by UGA.

Mentoring Experience (08 Undergraduates)

At UGA (mentoring):

- Kevin Truong (Georgia Southern University) NSF REU in Summer 2024
- Jiya Porwal (UGA student) in Summer 2026

Supervision of Undergraduate Student Research (Fall 2020)

- Adiba Mosharraf, B.Sc. student (Oceanography) at BMU, Bangladesh (Manuscript is Under review in a Q1 Journal)
- Md. Ashif Imam Khan, B.Sc. student (Oceanography) at BMU, Bangladesh (Research published in a Q1 Journal)
- Amanullah Rishat, B.Sc. student (Oceanography) at BMU, Bangladesh
- Saif Khan Sunny, B.Sc. student (Oceanography) at BMU, Bangladesh (Manuscript is Under review in a Q2 Journal)
- Two at Bangladesh Marine Academy, Bangladesh.

Other Engagements with Students:

- Worked as a course coordinator for 3rd year (Junior) undergraduate students (2019-2021)
- Facilitated field trips for the sessional work

Funded Grants

- Evaluation of Satellite Derived Ocean Chlorophyll-a Concentration in the Exclusive Economic Zone of Bangladesh, University Grant Commission Bangladesh, 2019-2021, \$16,519. Role: Co-PI.
- Spatio-temporal Variability of Stratification in the Northern Bay of Bengal, Ministry of Science and Technology, Bangladesh, 2018-2019, \$8,260. Role: Co-PI.

Applied Grants (not funded)

2025: NASA FINESST (\$98,524);

Title: New algorithms to estimate total suspended solids in the South Atlantic Bight.

2024: NASA FINESST (\$149,789);

Title: New algorithms to estimate total suspended solids in the South Atlantic Bight.

Individual Research Project/Thesis

- 2017: Important drivers for the massive bloom of *Mediopyxis helysia* Kühn at Helgoland Roads, Alfred Wegener Institute (AWI), Germany. (Individual research project as a requirement for NF-POGO graduation).
- 2015: Mesoscale Variability of Eddies-An Investigation into the Spatial and Temporal Scales in the Bay of Bengal, National Institute of Oceanography, India. (M.Sc. thesis).

Awards and Scholarships

2024: The Oceanographic Society Travel Award (\$1900) for presenting at the Ocean Optics meeting in Canary Islands in Spain.

2024: OCB (NASA & NSF) Travel Award (~\$400, housing and meals) to participate NASA PACE Hackweek 2024 at UMBC, MD.

2024: POGO - Alfred Wegener Institute Travel Awards (\$2000) to participate a workshop in Germany.

2022: IOCCG Training Award in France funded by OCB Travel Award (~\$2000).

2019: NSF Travel Award to participate Oceanhackweek 2019 in Seattle, WA (~\$800) and meals + housing.

2016: The Nippon Foundation-POGO Centre of Excellence Fellowship on Ocean Observation at AWI, Germany (~ \$20,000).

2015: Embassy of France in Bangladesh Research and Training Award (~\$1200).

2015: Dissertation Fellowship at National Institute of Oceanography, Goa, India (~\$1100).

2015: IFIC Bank Research Award at the University of Dhaka, Bangladesh (~\$300).

Peer-Reviewed Publications

Total: 24 articles, 248 citations in Google Scholar; 4 under review, 7 in preparation.

***Corresponding Author**

Scopus/WoS-indexed Journal Articles (14)

24. 2026: Barrett, B. S., Mahmud, N., **Masud-Ul-Alam, M.**, Ray, P., Habib, S., Foysal, M., Mili, M. I. J., & Biswas, M. H. (2026). Variability of tropical cyclogenesis in the Bay of Bengal across the phase space of the Madden–Julian Oscillation. *Atmospheric Science Letters*, 27(6), e70051. <https://doi.org/10.1002/asl2.70051>.
23. 2026: **Masud-Ul-Alam, M.**, Khan, M. A. I., & Turjo, K. E. Z. (2026). Inadequate in situ observations make the northern Bay of Bengal a blind spot for models. *Discover Oceans*, 3(1), 33. <https://doi.org/10.1007/s44289-026-00148-y>.

22. 2026: Torkelson, M., Bresnahan, P. J.*, Rivero-Calle, S., **Masud-Ul-Alam, M.**, Brewin, R. J. W., & Wells, D. (2026). Integrating in situ measurements and satellite imagery for coastal physical and biological analysis in the Cape Fear coastal region. *Remote Sensing*, 18(10), 1524. <https://doi.org/10.3390/rs18101524>.
21. 2025: Barrett, B. S.*, Davies, A. R., Spedero, C. P., Mahmud, N., Smith, J. P., **Masud-Ul-Alam, M.**, Biswas, M. H., & Ahmed, M. K. (2025). Subseasonal variability of ocean currents in the Bay of Bengal. *Discover Oceans*, 2(1), 1–14. <https://doi.org/10.1007/s44289-025-00081-6>.
20. 2025: Hossain, N. I., Rahman, S. M. M.*, Hossain, M. A., Hafiz, K. B., Sowrav, S. F. F., **Masud-Ul-Alam, M.**, & Choudhury, T. R. (2025). Decoding phytoplankton quality in the waterways of the Sundarbans mangrove forest of Bangladesh. *Marine Pollution Bulletin*, 213, 117669. <https://doi.org/10.1016/j.marpolbul.2025.117669>.
19. 2024: Khan, M. A. I., **Masud-Ul-Alam, M.***, Islam, M. N., Hossain, A., & Rahman, S. M. M. (2024). A model-based comparison of paleo-physical processes in the Bay of Bengal. *Modeling Earth Systems and Environment*, 10(6), 6659–6677. <https://doi.org/10.1007/s40808-024-02165-2>.
18. 2023: Fardoshi, T., Golder, M. R., Rouf, M. A.*, & **Masud-Ul-Alam, M.** (2023). Seasonal and spatial variability of chlorophyll-a in response to ENSO and ocean current in the maritime boundary of Bangladesh. *Journal of Marine Sciences*, 2023, 2843608. <https://doi.org/10.1155/2023/2843608>.
17. 2022: **Masud-Ul-Alam, M.***, Khan, M. A. I., Barrett, B. S., & Rivero-Calle, S. (2022). Surface temperature and salinity in the northern Bay of Bengal: In situ measurements compared with satellite observations and model output. *Journal of Applied Remote Sensing*, 16(1), 018502. <https://doi.org/10.1117/1.JRS.16.018502>.
16. 2022: **Masud-Ul-Alam, M.***, Khan, M. A. I., Barrett, B. S., Rivero-Calle, S., Golder, M. R., & Rouf, M. A. (2022). Spatial variability of the winter thermal inversion in the northern Bay of Bengal. *Regional Studies in Marine Science*, 52, 102417. <https://doi.org/10.1016/j.rsma.2022.102417>.
15. 2021: **Masud-Ul-Alam, M.***, Sarker, S., & Khan, M. A. I. (2021). The decadal response of vegetation in the Sundarbans mangrove forest to climate variabilities: Observing from space. *Remote Sensing of Earth System Sciences*. <https://doi.org/10.1007/s41976-021-00055-0>.
14. 2021: Mili, M. I. J.*, Ahmed, M. K., **Masud-Ul-Alam, M.**, Hasnain, M., Khan, M. A. I., Loodh, R., Hasan, A.-A., Hossain, K. B., & Nahian, S. A. (2021). In situ datasets of important physical and biochemical parameters on the continental shelf of the northern Bay of Bengal. *Data in Brief*, 35, 106947. <https://doi.org/10.1016/j.dib.2021.106947>.
13. 2020: **Masud-Ul-Alam, M.***, Khan, M. A. I., Islam, M. N., & Rahman, S. M. M. (2020). Modeling spatiotemporal variability of suspended matter and its relation with hydrodynamic parameters in the northern Bay of Bengal. *Modeling Earth Systems and Environment*. <https://doi.org/10.1007/s40808-020-01053-9>.

12. 2020: Sarker, S.*, **Masud-Ul-Alam, M.**, Hossain, M. S., Chowdhury, S., & Sharifuzzaman, S. M. (2020). A review of bioturbation and sediment organic geochemistry in mangroves. *Geological Journal*. <https://doi.org/10.1002/gj.3808>.
11. 2020: **Masud-Ul-Alam, M.***, Khan, M. A. I., Sunny, S. K., Rahman, M. A., Rahman, M. S., Mahmud, M. B., & Shaheen, M. A. R. (2020). An exclusive in situ dataset on physicochemical parameters in the gappy northern Bay of Bengal. *Data in Brief*, 32, 106024. <https://doi.org/10.1016/j.dib.2020.106024>.

Other Peer-reviewed Journal Articles (10)

10. **2026:** Hasan, M. A., Koly, F. A., Sowrav, S. F. F.*, Rahman, S. M. M., Alam, N., **Masud-Ul-Alam, M.**, Kabir, M. Z., Sarder, S. R., & Ahmed, K. (2026). Assessing the dynamics of water quality and their implications for the biochemical health of a river ecosystem: A remote-sensing-based study of the Shitalakshya River, Dhaka. *Bangladesh Maritime Journal*, 10(1). <https://doi.org/10.70279/bmj-v2-1124>.
9. 2025: Mili, M. I. J.*, Shaheen, M. A. R., **Masud-Ul-Alam, M.**, Reza, M. S., Mohammad, K., & Moontahab, A. (2025). Nutrients and chlorophyll-a in the northern Bay of Bengal: The need for focused research on Bangladesh's shelf. *Bangladesh Maritime Journal*, 9(1). <https://doi.org/10.70279/bmj-v8-1030>.
8. 2024: Loodh, R.*, Talokder, M. A. R., Rahman, M. S., **Masud-Ul-Alam, M.**, & Islam, M. T. (2024). Assessment of in-situ, satellite-, and model-derived physicochemical conditions around Saint Martin's Island, northeastern Bay of Bengal. *The Journal of NOAMI*, 38(1–2), 1–15.
7. **2023:** **Masud-Ul-Alam, M.***, Islam, M. A., Rahman, S. M. M., Khan, M. A. I., Hossain, M. A., Ibtehal, K. M., & Mosharraf, A. (2023). Variation of surface chlorophyll-a and its teleconnection to climatological factors in the Bay of Bengal. *Bangladesh Maritime Journal*, 7(1). <https://doi.org/10.70279/bmj-v7-i1-1053>.
6. 2023: Sunny, S. K.*, Firdaws, S. J., Zaman, M., & **Masud-Ul-Alam, M.** (2023). Commercially and economically important fish species from an otter-trawling vessel of the Bay of Bengal, Bangladesh. *Journal of the Asiatic Society of Bangladesh, Science*, 49(1), 13–25. <https://doi.org/10.3329/jasbs.v49i1.67592>.
5. 2020: **Masud-Ul-Alam, M.***, Chowdhury, S. M. B., Rashed-Un-Nabi, M., Alam, M. W., Khan, M. A. I., & Hafiz, K. B. (2020). Seasonality of stratification along the offshore area of the northern Bay of Bengal. *Bangladesh Maritime Journal*, 4(1). <https://doi.org/10.70279/bmj-v4-i1-1009>.
4. 2019: Kashem, M.*, Ahmed, M. K., Akhter, M. A. E., Nahian, S. A., Loodh, R., & **Masud-Ul-Alam, M.** (2019). Simulation of Tropical Cyclone Viyaru using the Weather Research and Forecasting (WRF-ARW) mesoscale model. *Journal of DEW-DROP*, 6(1). [Link](#)
3. 2019: Rahman, M. S., Ahmed, M. K., Chaudhuri, D., Loodh, R., Kashem, M., & **Masud-Ul-Alam, M.** (2019). Spatial and temporal analysis of satellite-derived sea-surface salinity in the northern Bay of Bengal. *Journal of DEW-DROP*, 6(1). [Link](#)
2. 2017: Chowdhury, K. M. A., Ahmed, M. K., Malik, M. K., **Masud-Ul-Alam, M.**, & Chowdhury, S. M. B. (2017). Interaction of sea-surface temperature and precipitation

during Indian Ocean Dipole years in the Bay of Bengal. *Bangladesh Maritime Journal*, 1(1). <https://doi.org/10.70279/bmj-v1-i1-1098>.

1. 2017: Chowdhury, K. M. A., Ahmed, M. K., **Masud-Ul-Alam, M.**, & Rani, S. (2017). Geostrophic current in the Bay of Bengal using satellite altimetry. *Journal of DEW-DROP*, 4(1). [Link](#)

Conference Paper

1. 2023: Mosharraf, A., **Masud-Ul-Alam, M.**, Begum, F., Dey, M., et al. Interaction of chlorophyll and sea-surface temperature of tropical cyclones with eddies in the Bay of Bengal. *9th International Conference on Water and Flood Management (ICWFM 2023)*, Dhaka, Bangladesh, October 2023. [Link](#)

Under review:

4. 2026: Mosharraf, A., **Masud-Ul-Alam, M.***, & Barrett, B. S. (2026). Interactions between tropical cyclones, oceanic eddies, and sea-surface parameters in the Bay of Bengal. *Journal of Marine Systems*.
3. **2026: Masud-Ul-Alam, M.***, Turjo, K. E. Z., Sarker, S., Ome, J. H., & Bithi, T. M. (2026). Ocean color remote sensing in the Bay of Bengal: A validation-first framework for the hyperspectral era. *Science of Remote Sensing*.
2. 2025: **Masud-Ul-Alam, M.***, & Khan, M. A. I. (2025). A comparison between satellite-derived chlorophyll-a and *in situ* observations in the northern Bay of Bengal. *Regional Studies in Marine Science*.
1. 2024: Sunny, S. K., **Masud-Ul-Alam, M.***, Khan, M. A. I., Mahmud, S. S., & Rahman, S. M. M. (2024). In situ observations of biophysicochemical parameters and their spatial distribution in the northeastern Bay of Bengal. *Transactions on Maritime Science*.

In preparation:

7. 2026: **Masud-Ul-Alam, M.**, & Rivero-Calle, S.* (2026). Long-term variability of MODIS-derived total suspended solids in the South Atlantic Bight. *Remote Sensing of Environment*. (First draft completed).
6. 2026: **Masud-Ul-Alam, M.**, Rivero-Calle, S.*, Bresnahan, P., Feldman, G. C., Holmes, A., Morrison, J., Hong, L., Scott, A., & Bailey, S. (2026). Performance analysis of SeaHawk compared with NASA and NOAA ocean-color satellites. *Remote Sensing of Environment*. (First draft completed).
5. 2026: **Masud-Ul-Alam, M.***, & Rivero-Calle, S.* (2026). A new global algorithm to estimate total suspended solids using PACE OCI data. *Remote Sensing of Environment*. (In review with co-author).
4. 2026: Rivero-Calle, S.*, Kolluru, S., **Masud-Ul-Alam, M.**, Lowin, B., Pellegrino, A., & Ohall, A. (2026). Ocean color satellite radiometry closure experiment in optically complex South Atlantic Bight waters. *Remote Sensing*. (In review with co-authors).
3. 2026: Dias Silva, F., **Masud-Ul-Alam, M.**, & Rivero-Calle, S.* (2026). Machine-learning automated detection of internal waves using SeaHawk and Sentinel-3 ocean-color imagery. *Remote Sensing*.

2. 2026: **Masud-Ul-Alam, M.***, Barrett, B. S., & Khan, M. A. I. (2026). A new machine-learning-based chlorophyll-a algorithm for Sentinel-3 OLCI in Case-2 waters of the northern Bay of Bengal. *Science of Remote Sensing*. (First draft completed).
1. 2026: Lang, S., Omand, M., Jones-Kellett, A. E., **Masud-Ul-Alam, M.**, Snow, T., Cornec, M., & Puppa Kocsis, V. (2026). Biogeochemical variability in Gulf Stream eddies using satellites and BGC-Argo. *Geophysical Research Letters*.

Scientific/Conference Presentations

Presenter is underlined

16. 2026: **Md Masud-Ul-Alam** & Sara Rivero-Calle (2026, December). Satellite-derived long-term variability and forecasting of total suspended solids in the South Atlantic Bight leveraging 24 years of MODIS-Aqua data. *AGU Fall Meeting 2026*, San Francisco, CA, United States. Submitted. [Link](#)
15. 2026: **Md Masud-Ul-Alam** & **Md Ashif Imam Khan** (2026, December). In situ assessment of underwater light regimes and phytoplankton productivity across Kuakata and the Sundarbans. *AGU Fall Meeting 2026*, San Francisco, CA, United States. Submitted. [Link](#)
14. 2024: **Md Masud-Ul-Alam** & Sara Rivero-Calle (2024, November). Building a new regional algorithm to estimate total suspended solids in the South Atlantic Bight. *Gray's Reef National Marine Sanctuary Science Symposium*, Savannah, GA, United States. [Poster] [Link](#)
13. 2024: **Md Masud-Ul-Alam** & Sara Rivero-Calle (2024, October). Building a new regional algorithm to estimate total suspended solids in the South Atlantic Bight. *Ocean Optics XXVI Conference*, Las Palmas de Gran Canaria, Spain. [Poster] [Link](#)
12. 2024: **S. Rivero-Calle**, F. Dias Silva, **M. Masud-Ul-Alam**, B. Lowin, Z. Ljubešić, A. Matek, H. Mihanović, H. Čížmek, & B. Čolić (2024, October). Chasing internal waves with ocean-color methods. *Ocean Optics XXVI Conference*, Las Palmas de Gran Canaria, Spain. [Poster] [Link](#)
11. 2024: **Md Masud-Ul-Alam** (2024, May). Performance analysis of SeaHawk with other NASA ocean-color satellites. *Marine Sciences Student Symposium, University of Georgia*, Georgia, United States. [Oral] [Link](#)
10. 2024: **S. Rivero-Calle**, S. Kolluru, **Md Masud-Ul-Alam**, B. Lowin, M. Hunt, A. Ohall, & F. J. Dias Silva (2024, February). Using high-resolution ocean optics to better constrain ranges of marine phytoplankton variability. *Ocean Sciences Meeting 2024*, New Orleans, LA, United States. [Oral] [Link](#)
9. 2023: **Md Masud-Ul-Alam**, S. Rivero-Calle, G. C. Feldman, A. Holmes, J. Morrison, L. Hong, A. Scott, P. Bresnahan, & S. Bailey (2023, November). SeaHawk low-cost ocean-color CubeSat produces high-spatial-resolution and high-quality data: A comparison with NOAA VIIRS and NASA MODIS Terra and Aqua. *International Ocean Colour Science Meeting*, St. Petersburg, FL, United States. [Poster] [Link](#)
8. 2023: **S. Rivero-Calle**, S. Kolluru, **Md Masud-Ul-Alam**, B. Lowin, M. Hunt, A. Ohall, & F. J. Dias Silva (2023, November). Unveiling submesoscale processes with high-resolution ocean optics. *International Ocean Colour Science Meeting*, St. Petersburg, FL, United States. [Poster] [Link](#)
7. 2023: **S. Rivero-Calle**, J. Morrison, G. C. Feldman, A. Holmes, P. Bresnahan, **M. Masud-Ul-Alam**, B. Lowin, S. Bailey, A. Scott, & L. Hong (2023, June). The potential of ocean-color CubeSats for operational satellite oceanography: The SeaHawk-HawkEye example.

Third International Operational Satellite Oceanography Symposium, Busan, South Korea. [Oral] [Link](#)

6. **2023: M. Masud-Ul-Alam**, S. Rivero-Calle, G. C. Feldman, A. Holmes, J. Morrison, L. Hong, A. Scott, P. Bresnahan, & S. Bailey (2023, 4-9 June). SeaHawk low-cost ocean-color CubeSat produces high-spatial-resolution and high-quality data: A comparison with NOAA VIIRS and NASA MODIS Terra and Aqua. *ASLO Aquatic Sciences Meeting 2023*, Palma de Mallorca, Spain. [Oral] [Link](#)
5. 2022: **S. Rivero-Calle**, J. Morrison, G. C. Feldman, A. Holmes, S. Bailey, **M. Masud-Ul-Alam**, M. Hunt, B. Lowin, & P. Bresnahan (2022, 2-7 October). A CubeSat, a ship, and a zoo of algae: Three new optical-oceanography applications at the Skidaway Institute of Oceanography. *Ocean Optics XXV Conference*, Quy Nhon, Vietnam. [Oral] [Link](#)
4. 2022: **S. Rivero-Calle**, J. Morrison, G. C. Feldman, A. Holmes, S. Bailey, & **M. Masud-Ul-Alam** (2022, 24 February - 4 March). Increasing the resolution of ocean observations while reducing costs: CubeSats and ship flow-through systems. *Ocean Sciences Meeting 2022*, Online. [Oral] [Link](#)
3. 2022: **S. K. Sunny*** & **M. Masud-Ul-Alam** (2022, October). Association of photosynthetic pigments with oceanic environmental parameters in the northeastern Bay of Bengal. *International Conference of Physical Sciences*, Shahjalal University of Science and Technology, Sylhet, Bangladesh. [Oral]
2. 2022: Mosharraf, A., & **Masud-Ul-Alam, M.*** (2022, November). Upper ocean responses between oceanic eddies and tropical cyclones in the Bay of Bengal [Poster presentation]. *2021–2030 UN Decade of Ocean Science: Imperative for Bangladesh International Seminar*, Bangladesh Maritime University, Bangladesh. [Link](#)
1. 2020: **M. Masud-Ul-Alam**, M. K. Ahmed, & S. P. Kumar (2020, January). Mesoscale variability of eddies: An investigation of spatial and temporal scales in the Bay of Bengal. *International Conference on Earth and Environmental Sciences and Technology for Sustainable Development*, Dhaka, Bangladesh. [Oral]

Selected Specialized Training and Workshops

- 2026: NASA ARSET training on Monitoring Water Quality in Lakes and Coastal Regions Using STREAM (online), 10 and 17 February 2026. [Link](#)
- 2024: Machine Learning for Ocean Studies at the Ocean Optics XXVI Conference, Las Palmas de Gran Canaria, Spain, 6–11 October 2024 (4-hour hands-on training). [Link](#)
- 2024: NASA PACE Hackweek, University of Maryland, Baltimore County, Baltimore, Maryland, USA, 4–8 August 2024. [Link](#)
- 2024: Communicating Science and Building Low-Cost Sensors for Ocean Observations Workshop, Alfred Wegener Institute, Germany, 2024.
- 2023: International Workshop on Western Boundary Current–Subtropical Continental Shelf Interactions, Savannah, Georgia, USA, 22–24 May 2023. [Link](#)
- 2023: Copernicus Marine Service Training Workshop for the North-West Shelf Seas Region (online), 24–25 January 2023. [Link](#)
- 2023: Programming with Python workshop, University of Georgia (online), 6–7 March 2023. [Link](#)

- 2022: IOCCG Summer Lecture Series, Frontiers in Ocean Optics and Ocean Colour Science, Institut de la Mer de Villefranche, Villefranche-sur-Mer, France, 18–29 July 2022. [Link](#)
- 2021: Virtual Workshop on the New Global Ocean Biogeochemistry Array, Building a Community of Biogeochemistry Float Data Users, 28–30 June 2021. [Link](#)
- 2019: Marine Resources Management and Blue Economic Development for Bangladesh, Fujian Institute of Oceanography, China, 2019 (21 days).
- 2018: Hands-on Training on CTD Operation, Data Collection and Analysis, Bangladesh Oceanographic Research Institute, Bangladesh, 2018 (3 days; funded by Hydro-Bios, Germany).
- 2018: Scientific Programming in Oceanography: R, MATLAB, Python, and Fortran, Bangladesh Oceanographic Research Institute, Bangladesh, 2018 (14 days).
- 2018: China–Bangladesh Cooperative Workshop on Ocean Forecasting and Remote Sensing in Marine Science, National Marine Environmental Forecasting Center, Beijing, China, 2018 (7 days).
- 2017: Coastal Summer School, How to Govern the Marine Environment: Baltic Sea and Sediment Services as a Case Study, Germany, 10–21 September 2017. [Link](#)
- 2016: Compact Course in Marine Science: Biogeochemical Oceanography and Climate Science, Alfred Wegener Institute, Germany, 2016 (4 days).
- 2015: Oceanographic Data Processing, Analysis, and Interpretation, National Institute of Oceanography, Goa, India, 2015 (3 months).

Citable Data

- 2020: Most Israt Jahan Mili, Shaheen Md. ashifur Rahman, **Md Masud-Ul-Alam**, Reza Md. Shafiq, Billah Md. Mohtasim, Kayes Mohammad (2020), “Nutrients and Chlorophyll-a in-Situ Dataset Collected during 2020 in Bangladesh”, *SEANOE*, doi: [10.17882/98184](https://doi.org/10.17882/98184).
- 2020: **Masud-Ul-Alam, Md***; Sunny, Saif Khan; Rahman, Md Atiqur; Shaheen, Md. Ashifur Rahman; Mahmud, Md. Bayzid; Khan, Md. Ashif Imam (2020), “Invaluable CTD and Mixed Layer Depth data in northern coast of the Bay of Bengal”, *Mendeley Data*, V2, doi: [10.17632/v8gnfjgwmr.2](https://doi.org/10.17632/v8gnfjgwmr.2).

Sea-going and Fieldwork Experience (>57 days)

- 12/2025: RV Savannah, suspended sediments sample collection, radiometric measurements and satellite match-up. (2 days).
- 04/2025: RV Savannah, P.I. Dr. Sara Rivero-Calle. I was responsible for collecting water samples, coordinating CTD, HyperPro, and SVC operation and training 10 graduate students on sample processing and optical oceanography equipment. (3 days).
- 09/2024: RV Savannah, radiometric measurements and satellite match-up. (3-days).
- 07/2023: Field campaign in Croatian waters, CTD, SVC, satellite match-up; P.I. Dr. Zrinka Ljubesic (14 days).
- 01/2022: RV Shearwater, tracking fronts using dyes, HyperPro, and Aircraft with the research team at Duke University (5 days).

09/2021: RV Savannah, a day cruise for glider deployment, CTD, sediment and water sample in the SAB.

02/2020: BNS Turag (with the collaboration of Bangladesh Navy), Sediment sample collection, CTD, chlorophyll and nutrients measurements in the northern Bay of Bengal as a Co-PI (7-days).

02/2018: BNS Kapatakhyia (with the collaboration of Bangladesh Navy), Hydrographic parameters measurements in the Bay of Bengal (5 days).

10/2017: RV Elisabeth Mann Borgese, focused on coastal research in the Baltic Sea (2 days).

04/2017: RV Heincke, Special cruise including CTD measurement in the North Sea (5 days).

10/2016: RV Uthörn, Hydrographic parameters measurements in the North Sea (several 1-day cruises) with AWI.

11/2016: RV Mya II, Hydroacoustics and sediment sampling in the Wadden Sea (2 days).

02/2015: BNS Kapatakhyia (with the collaboration of Bangladesh Navy), Hydrographic parameters measurements, Niskin sampler, and plankton net operation in the Bay of Bengal (7 days).

In addition, I participated more than 10 fieldworks (~15 days) in the coastal and inland waters, as in the SAB a graduate student, and as a Co-PI in the Bay of Bengal.

Professional Service

To the Scientific Community:

- **Manuscript Reviewer (invited):** *Scientific Reports, Journal of Operational Oceanography* (2), *International Journal of Applied Earth Observation and Geoinformation, Physics and Chemistry of the Earth, Frontiers in Marine Science, Estuarine, Coastal and Shelf Science* (3), *Ocean & Coastal Management, Marine Pollution Bulletin, Ocean Dynamics, Journal of Electronic Imaging, Regional Studies in Marine Science* (9), *Environmental Monitoring and Assessment* (4), *Water, Air, & Soil Pollution, Aquatic Ecology, Watershed Ecology and the Environment*, and *Journal of the Marine Biological Association of the United Kingdom*. (Total **30 manuscripts**)
- **Proposal Reviewer:** Reviewer of NASA and NSF proposals submitted from my Lab (PI: Dr. Sara Rivero-Calle).

Leadership Experience at UGA Skidaway Institute of Oceanography and BMU

- Skidaway Marine Science Day volunteer in 2023, 2024, and 2025
- Student volunteer at Professional Industry Networking Event at UGA, 2025
- Member of Organizing Committee of *Marine Sciences Student Symposium* in 2024
- Volunteer at International Workshop on Western Boundary Current, 22-24 May 2023
- Volunteer at ASLO 2023 Aquatic Sciences Meeting in Palma, Spain
- Member of the BS in Oceanography Exam committee.
- Beyond the classroom, I was involved in setting up the biological and physical oceanography labs and curriculum revision for the BS in Oceanography program.

Specialized Skills

- Able to analyze geospatial, oceanographic, and satellite data and synthesize results into reproducible research products.
- Advanced knowledge of remote-sensing and geospatial tools, including ArcGIS, QGIS, SNAP, SeaDAS, ACOLITE, Google Earth Engine, ODV, Surfer-8, GitHub, JupyterHub, cloud computing, and LaTeX.
- Expert in **MATLAB, R, Python, and use of High-Performance Computing (HPC)**, applied extensively in M.S. and Ph.D. research.
- Experience using **machine learning, AI, and Agentic AI in research; GitHub, CDO, and Unix/Bash scripting.**
- Experience in leading and planning any inshore and offshore field campaign and operating oceanographic instruments (**CTD, Rosset sampler, Niskin, Grab Sampler, HyperPro, SVC, ACs, Fluorometer**, etc.).
- Experience of using microscope, spectrophotometer, fluorometer, DNA and RNA extraction, PCR etc.

Professional Affiliation

1. Student member of The Oceanographic Society, Rockville, MD, USA
2. Member of Association for the Sciences of Limnology and Oceanography, Washington, DC, 20036, USA
3. Student Member of American Geophysical Union
4. Member of NF-POGO Alumni Network for the Ocean
5. International Indian Ocean Expedition (IIOE-2) Early Career Scientist Network

References

1. **Dr. Sara Rivero-Calle**

Associate Professor

Skidaway Institute of Oceanography, University of Georgia

Email: rivero@uga.edu

Relationship: Ph.D. advisor and research supervisor at the University of Georgia.

2. **Dr. Clark Alexander**

Director and Professor

Skidaway Institute of Oceanography, University of Georgia

Email: Clark.Alexander@skio.uga.edu

Relationship: Ph.D. Committee Member, familiar with my research at UGA.

3. **Dr. Deepak Mishra**

Merle C. Prunty, Jr. Professor & Head

Director, [Small Satellite Research Lab](#)

Director, [Center for Geospatial Research](#)

Department of Geography, University of Georgia, Athens

210 Field Street, Room 209A, Athens, GA 30602

Email: dmishra@uga.edu;

Relationship: Ph.D. Committee Member and familiar with my research at UGA.

4. **Dr. Liang Hong**

Aerospace Engineer
NASA Goddard Space Flight Center
8800 Greenbelt, MD 20771, USA

Relationship: Research collaborator and coordinated with him in processing image requests and scheduling them as a part of UGA science team for SeaHawk mission.

5. **Dr. Philip J. Bresnahan**

Associate Professor
Department of Earth and Ocean Sciences
Center for Marine Science
University of North Carolina Wilmington
Email: bresnahanp@uncw.edu;

Relationship: Research collaborator familiar with my ocean-color remote-sensing, satellite-data evaluation, field observations, and collaborative publication work.